Universität Siegen

Naturwissenschaftlich-Technische Fakultät

Ubiquitous Computing

Privacy-aware robot reminders

Watch-mediated consent

Master’s thesis midterm

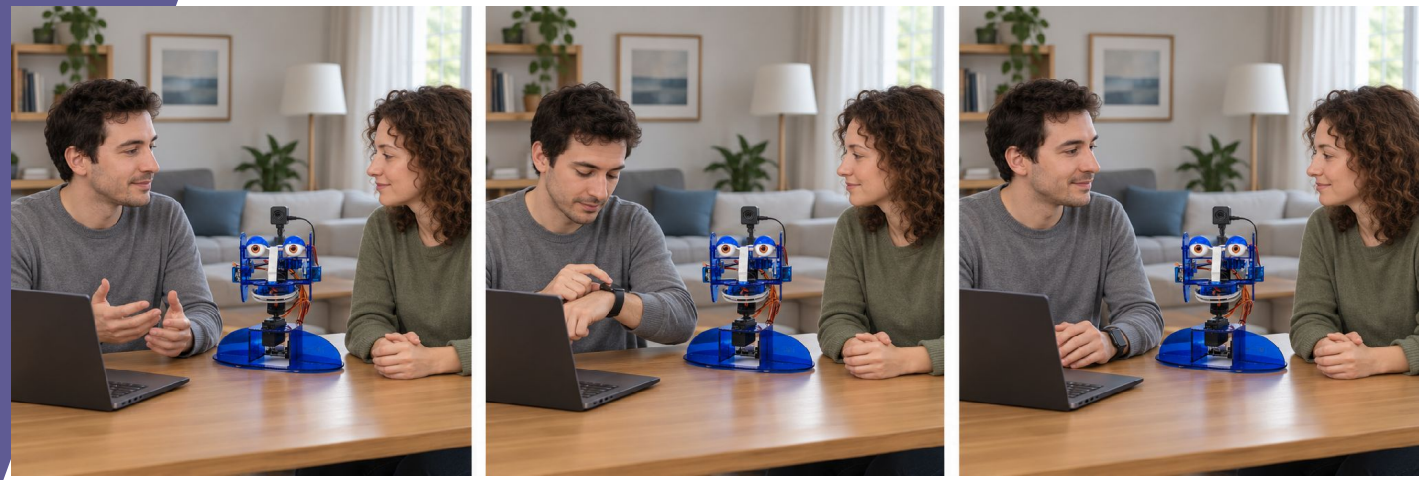


Photo courtesy ChatGPT

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The privacy problem

A reminder changes meaning when someone else can hear it

“Your doctor’s appointment is at 3 PM.”

Helpful when alone. Potentially revealing with a visitor.

The robot needs to choose public or private delivery.

The wearer makes the disclosure decision on the watch.

Research questions

Proposed questions for the current thesis scope

RQ1: How can a robot coordinate sensing and private consent?

RQ2: How acceptable are microphone, camera and fallback packages?

RQ3: When do users prefer memory or fresh consent?

System components

Ohbot and Bangle.js communicate through a Python host

Ohbot: spoken reminder or neutral greeting over USB.

Bangle.js: private Yes/No prompt and reminder over BLE.

Python: scheduling, sensitivity, recognition and policy.

Camera and microphone: estimate bystander presence.

Reminder delivery policy

Sensitivity and social context determine the delivery path

Non-sensitive: speak at the scheduled time.

Sensitive, no detected bystander: allow spoken delivery.

Bystander detected: request or retrieve a watch decision.

Yes: speak. No or no reply: deliver privately to the watch.

Sensing packages

Coverage and sensing burden differ across packages

Microphone: detects visitors who speak while listening.

Camera: detects visible faces, including silent visitors.

Fallback: microphone first, camera after an audio miss.

Face and voice identities remain separate.

Consent memory

Remembered choices reduce prompts but may become stale

Remember: reuse a recognized bystander’s explicit Yes/No.

Re-ask: request a fresh decision each time.

Both policies retain recognition galleries.

Expiry, topic-specific consent and watch editing are proposed.

Prototype progress

The repository contains the full delivery workflow

Implemented: reminder capture, scheduling and sensitivity.

Implemented: enrollment, sensing and watch consent.

Implemented: standalone and fused engines with both policies.

Prepared: presentation UI, benchmarks and survey materials.

Early technical measurements

Bundled CPU measurements support further testing

YuNet detection: 5.15 ms median per 640×480 frame.

SFace recognition: 7.33 ms per 112×112 face crop.

Resemblyzer: 23.74 ms per 6-second utterance.

Accuracy and end-to-end delay remain to measure.

Main study design

Six matched stories compare three sensing packages

Each person sees 2 microphone, 2 camera and 2 fallback stories.

Three topics crossed with friend/home and colleague/office.

Each package has one Yes/aloud and one No/private outcome.

Allocation and story order rotate across 36 cells.

Outcomes and analysis

Acceptability is the primary outcome

Primary: Q10 acceptability, fallback versus each single sensor.

Secondary: privacy, trust, sensing comfort and control.

Planned: ordinal mixed model for repeated participant ratings.

Power simulation and a cognitive pilot precede recruitment.

Study interpretation

The main outcome concerns described sensing packages

Main visitors speak, so the fallback camera stays inactive.

Timing and head movement differ between packages.

The no-sensor story changes policy and is exploratory.

Memory questions measure preference, not treatment effects.

Remaining work

Proposed sequence for the second half of the thesis

1. Freeze behavior, stimuli and approved study information.
2. Pilot comprehension and burden, then finalize power.
3. Recruit, collect ratings and test recognition robustness.
4. Analyze effects and limitations, then complete the thesis.

Midterm takeaway

The prototype is implemented. Human evaluation is next.

The watch provides private consent and delivery.

Early latency measurements support component feasibility.

The revised study separates sensing from memory preferences.

Feedback: retain this scope or add live fallback evaluation?

Discussion

Thank you

Study scope and fallback evaluation

Consent expiry and changing social context

Recognition robustness and sensing permissions